

Business Intelligence

ITB's customers are loyal and active, but their financial health rises and falls with their own spending pace. Using a year of card data, we map who is thriving, who is stretched and who is drifting away, and propose non-punitive ways to help each group spend within their means.

EXECUTIVE SUMMARY

Situation	The lender serves about 1.0M customers across contracts, applications, leads and calls. Credit and sales decisions both depend on how customers behave, but each team sees only part of that history.
Complication	Two questions are answered separately: can this customer repay, and will this customer buy again? Sales effort is spread evenly, so a healthy customer may be dormant while an engaged customer may be risky.
Key Findings	• Financial health comes mainly from credit behaviour, such as trust from other lenders and repayment history, rather than income. • Engagement comes from how often a customer signs and applies, and the score ranks likely buyers far better than simple RFM. • Most of the expected value sits in a small group of healthy, active customers, and the best moment to offer credit is when a loan is ending.
Question	How can the lender measure financial health and engagement with scores managers can explain, and turn them into one clear action per customer?
Answer	We combined the two scores into nine named segments, then applied rules in priority order: block risk first, then act by segment, then adjust product, channel and timing to each customer's behaviour. Every customer receives one action with a plain-language reason. We recommend an A/B test with control groups before wide rollout, and regular recalibration of the scores

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Data Preprocessing

Raw data was cleaned into one reliable, leakage-free base for customer scoring.

Data issues	
CONTRACT	• 1.95M duplicated contracts from a profile join • XNA, 0 used as missing values; income outliers up to 9B VND • AMT_CREDIT = 1 placeholder (345 contracts); TRANSAC holds dates, not channel codes
CUSTOMER	• XNA = missing (education 16%, salary frequency 99%); FLAG_CAR_OWNER is constant • Implausible values: age 2–124, 99 children; inconsistent province names
LEADS	• 39K exact duplicates; • sentinel date 3000-01-01 on 27% of leads; • MAX_OFFER as low as 9 VND
PAST_BEHAVIOR	• ACTION mixes channel and event type
CALL	• 12.5K non-unique CALL_ID; 248 raw response codes; 74% of calls fall after the snapshot (data runs to 08/2026)

- Removed the duplicated records. Exact duplicate leads were also dropped (39K).

- Placeholder and implausible entries were turned into missing values and each was flagged. This keeps the original information visible and avoids assumptions that could bias the scores. Product codes, provinces and call outcomes were also unified

- Derived signals that reflect customer behavior: overdue history, collection calls, lead outcomes, and channel. The snapshot date is fixed at 04/06/2025, only earlier data is used for scoring. Later calls are kept to validate the scores against real outcomes.

Context and Data

The target customer base is a Tet season cohort: all 1,000,528 customers submitted applications between December 2024 and January 2025. The data cutoff date is June 4, 2025.

Each customer has signed or applied for **4.3** loan contracts on average, so the customer base has a deep credit history.

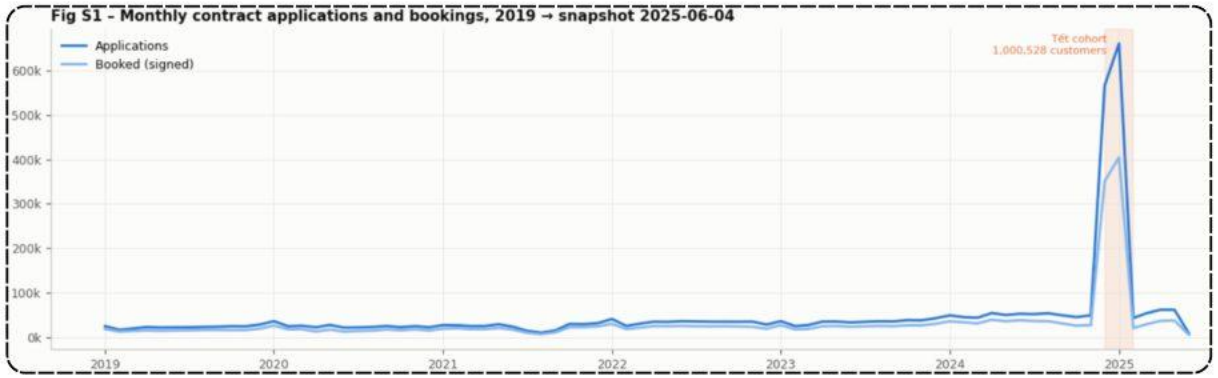
All customers applied for a loan during Tết (Dec 2024 – Jan 2025), so the base is one seasonal cohort.

64.5% of customers have expressed interest in a new loan through the app, website, sales agents or partner stores, about **3.3** times each

Only **47%** of customers have checked their loan eligibility or pre-approved limit themselves, but those who did checked it about **9** times each.

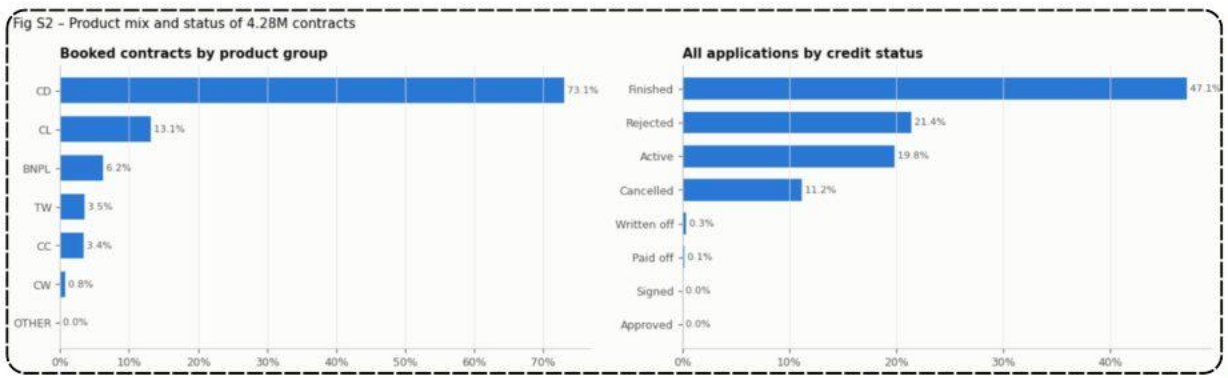
Only **36%** of customers have ever been contacted by phone by the sales team, yet those reached received about 11 calls each.

Scale: 2.89 million disbursed contracts, total volume of approximately VND 30.8 trillion; 849 thousand active contracts.



Insight (Fig S1)

The monthly application volume surged to a**round 570k–660k** in Dec 2024 – Jan 2025, approximately **12 times higher** than the normal rate (~50k/month). Prior to this, monthly applications **grew steadily** from ~20k/month (2019) to ~50k/month (2024).



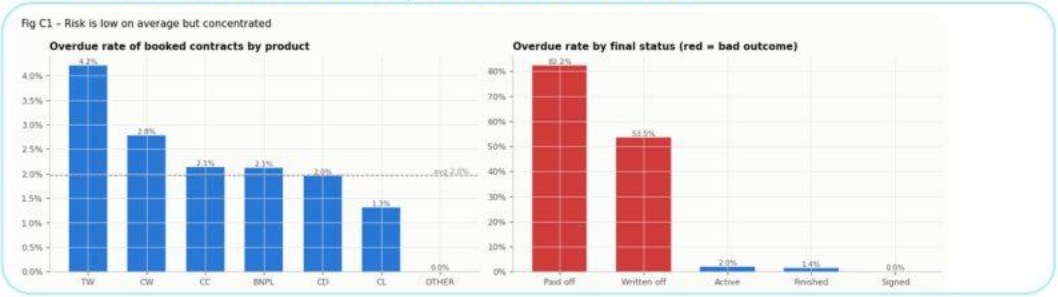
Insight (Fig S2)

- **Consumer Durable Loans** (CD) account for 73% of contracts, characterized by small ticket sizes (median: VND 5.3 million).
- **Cash Loans (CL)** account for only 13% of total contracts but match CD in value (VND 11.962 trillion vs. VND 13.786 trillion; median: VND 21 million).
- **Across all applications:** approximately 21% were rejected and 11% were cancelled, meaning 1 out of every 3 applications failed to convert into a contract.

Key Takeaway

Consumer Durable Loans (CD) serve as the primary acquisition channel for acquiring new customers, while Cash Loans (CL) drive value generation. The CD $\rightarrow$ CL / CC cross-selling pipeline represents the main growth lever.

Low to moderate risk, but concentrated

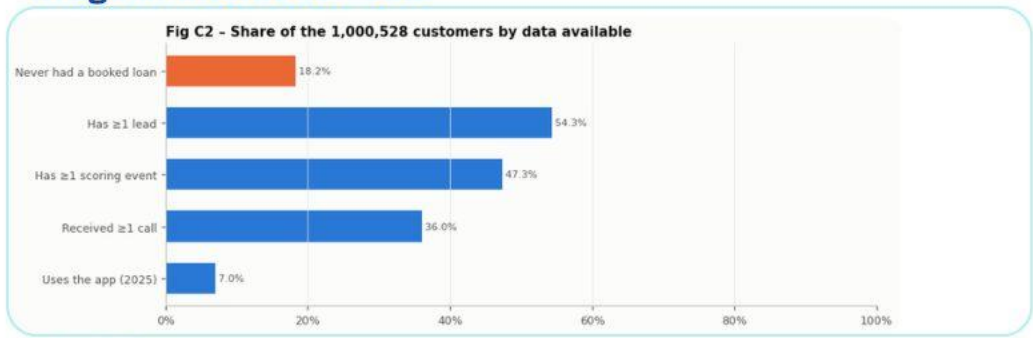


Insight: The average delinquency rate is only about 2% per contract; however, **6.2% of customers (50,991 individuals) have been delinquent at least once.**

- By product: The rate for motorcycles (TW) **is 4.2%, three times higher than that for cash loans** (CL), which stands at 1.3%.
- By status: 50–80% of contracts categorized as "Written off," "Sold," or "Paid off" have a history of delinquency, whereas only 1–2% of "Finished" or "Active" contracts do.

Implication: Delinquency flags (DUE) serve as reliable risk indicators. Since risk is not distributed uniformly, a customer-level scoring approach rather than one based solely on product type is necessary.

Irregular interaction data



Insight:

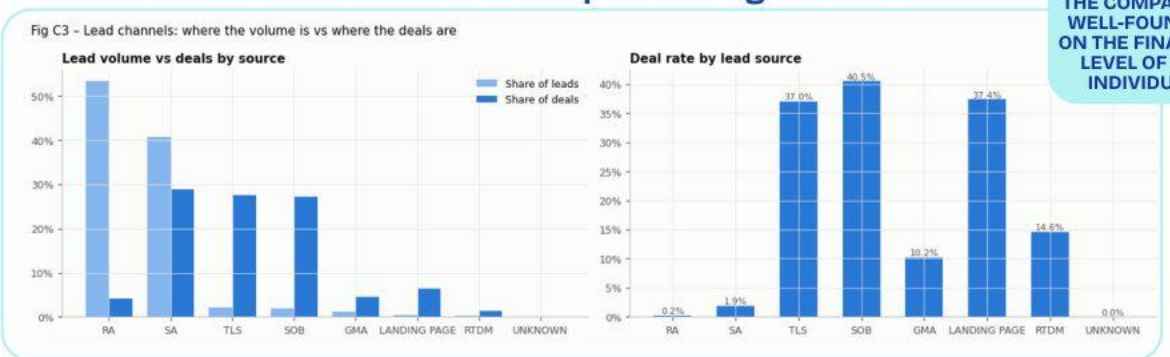
- **18.2% of customers (182,000) have never received a loan disbursement**.
- Among the customer base, 64.5% have an associated lead record, 47% have a scoring event, 36% have received a phone call, and **only 7% have used the app in 2025**.

Implication: It is not appropriate to score all customers using the same methodology. Customers with no prior history should be categorized with a specific label: "No credit history." The engagement scoring model must be robust enough to handle missing data. For instance, if a customer has not yet been contacted, the "phone call" component should be excluded from the formula rather than being assigned a score of zero.

PROBLEMS STATEMENT

THE COMPANY LACKS A CLEAR, WELL-FOUNDED PERSPECTIVE ON THE FINANCIAL HEALTH AND LEVEL OF ENGAGEMENT OF INDIVIDUAL CUSTOMERS.

Sales resource allocation to low-performing channels

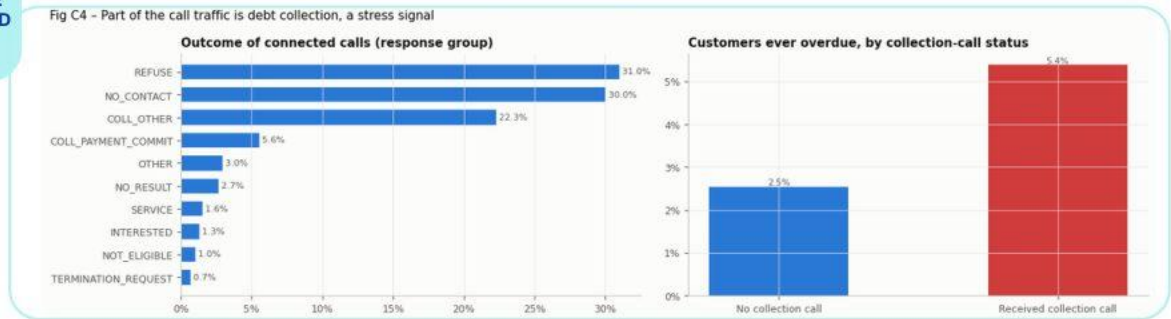


Insight:

- **RA generates 53% of total leads but accounts for only 4% of deals** (0.2% conversion rate).
- **TLS, SOB, and Landing Pages represent just 4.3% of leads, yet drive 61% of deals** (37–40% conversion rate).
- SA falls in the middle: 41% of leads and 29% of deals.

Implication: Telesales resources must be prioritized toward prospects exhibiting high engagement signals. The Engagement Score in Challenge 2 serves as the primary mechanism to filter and target these high-intent leads.

A fraction of call volume is attributed to debt collection, indicating underlying operational friction.



Insight:

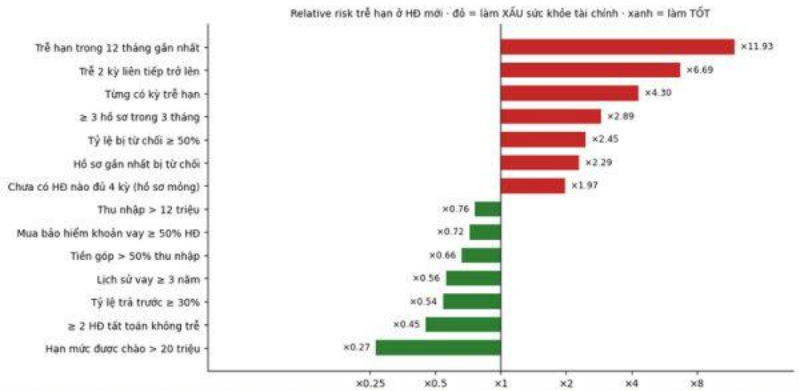
- **Debt collection calls account for 4.4% of total call volume**, targeting approximately 40,000 customers. The connection rate stands at 85%, compared to just 10% for sales calls.
- Customers who received debt collection calls have a past-due rate of **5.4%, roughly 2.1 times higher than those who did not (2.5%)**.
- Among connected calls: 31% were rejections, 30% failed to reach the intended person, and 28% were debt collection responses.

Implication: Debt collection calls are categorized under Financial Health and excluded from the Engagement metric. Otherwise, customers who "answer frequently" due to collection efforts would be misclassified as highly engaged.

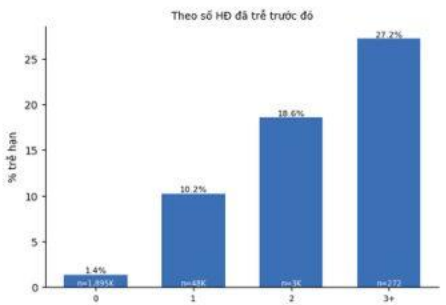
Customer Financial Health Score

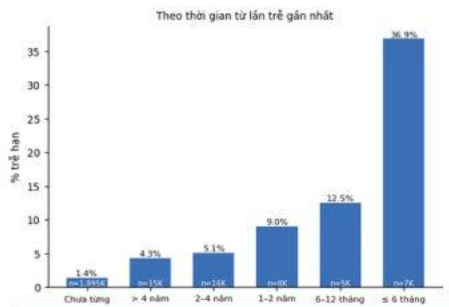
Past repayment behavior and lender assessment separate good from risky customers most clearly

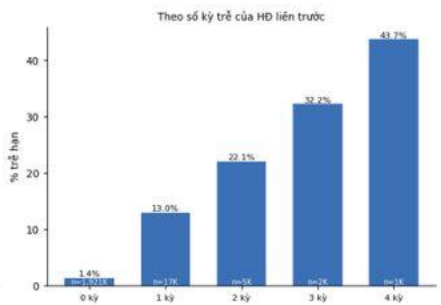
Past lateness and lender rejection are the strongest risk signals



- Late in the last 12 months: 11.9x; 2 consecutive late installments: 6.7x; 50% or more rejected: 2.5x
- Lender trust and proven on-time contracts are the strongest protection.
- Risk repeats and fades slowly: one earlier late contract lifts the next contract's late rate from 1.4% to 10.2%, a late payment within 6 months means 36.9%, and it is still 4.3% after 4 years.







Lateness repeats and fades slowly: no earlier late contract 1.4%, one 10.2%, two 18.6%, three or more 27.2%. A late payment within 6 months means 36.9%, and it is still 4.3% after 4 years. 50.6% of contracts late at installment 1 are still late at installment 4.

Factor group	Evidence	Effect
Repayment history	Last late within 6 months 36.9%, fading to 4.3% after 4 years	Strong negative
Borrowing pressure and lender trust	Over 75% of applications rejected: 23.1% ever late (0% rejected: 4.1%); offered limit up to 5M 7.0%, above 50M 2.0%, never offered 10.6%	Strong
Credit track record	First contract 3.0% late, falling to 0.7% after 6 or more on-time contracts	Positive
Financial buffer	Down payment up to 20%: 2.8%; above 60%: 1.2%	Positive
Debt burden	Installment-to-income up to 35% about 2.0%; 35-50% 2.8%; above 50% 4.2%	Moderate, threshold effect
Income and stability	Income above 20M 1.8% vs 2.5% up to 5M; no declared income 4.3%	Weak
Loan insurance	Opposite direction at contract and customer level (self-selection)	Mixed, no direction imposed

Customer Financial Health Score

Behavioral indicators built only from data known before each cut-off date

Snapshot

- One row per customer per cut-off
- 5 cut-offs (Feb–Oct 2024)
- Scoring date 05/06/2025

Aggregate

- Contracts, applications, offers → customer level
- Windows: 1, 3, 6, 12, 24 months, 3 years
- Level and recent trend

Leakage guard

- Late flags only for contracts ≥125 days old
- Label starts after cut-off
- Status, demographics excluded

Quality check

- Zero denominator kept missing, not 0
- 23 invalid values set to missing
- 12 correlated pairs, 1 dropped

Constrain

- Direction per indicator: +1 / −1 / free
- Keeps score consistent with business logic
- Free for threshold effects

Model feature group	
Repayment history (16)	Late installments, on-time rate, days since last late
Lender trust & pressure (19)	Refusal ratio, offered limit, recent applications
Financial buffer (5)	Down payment ratio, largest down payment
Debt burden (5)	Active contracts, installment-to-income
Income & stability (8)	Declared income, income variability
Income-expense balance (1)	Income left after installments
Planning & protection (2)	Share of contracts with loan insurance

Feature selection

Redundancy check	Spearman and Pearson between indicators	12 pairs above 0.95; 1 dropped (above 0.99 on both)
SHAP screening	Drop indicators with SHAP share below 0.5%	55 candidates → 35 kept
Direction rules	Monotone constraints on repayment and rejection indicators	No indicator moves against business logic
Group test	Remove one group, measure AUC loss on validation	Lender trust −0.0199; repayment history −0.0151; others ~0

- Not used as inputs: demographics, contract status (paid-off contracts were late 82.1% of the time), PAST_BEHAVIOR (reserved for Engagement), CALL (starts after all training cut-offs).
- Data completeness at scoring date: 52.7% full history · 26.3% partial · 21.0% thin file → band N.
- Reading: score is richer on behavior than on capacity; income, debt and balance add almost no signal.

Customer Financial Health Score

Past repayment behavior and lender assessment separate good from risky customers most clearly

Model

LightGBM + TreeSHAP explanation

Metrics

AUC · Gini · KS

Input variables

Lender trust & pressure:
- offered limit,
- rejection ratio,
- last application rejected

Repayment history:
- late rate of last 3 contracts,
- on-time installments

Financial buffer: down payment ratio, down payment-to-income

Debt burden: installment-to-income, revolving contracts

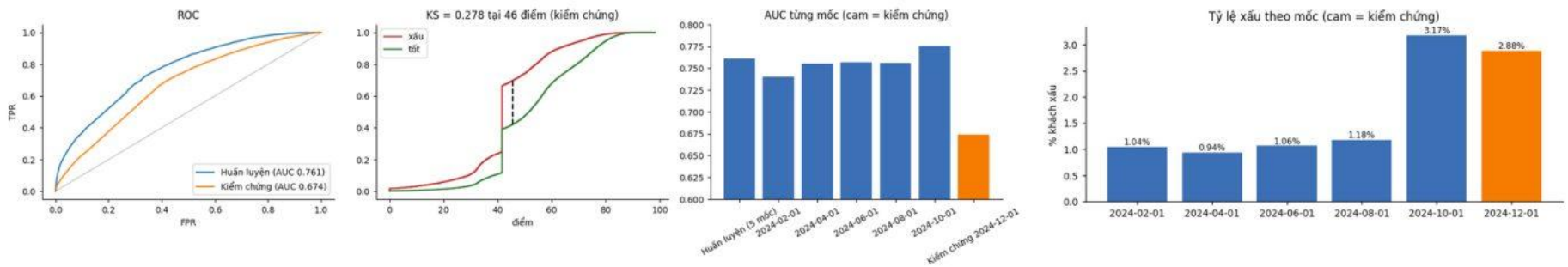
Income & stability: income variability, years of income declared

Training design

Label: late in first 4 installments or written off, within 2 months after cut-off

Maturity ≥125 days

Step 1: Snapshot at 5 cut-offs (02-10/2024) → Step 2: Train on 4, select on 10/2024 → Step 3: Test on 12/2024 (never used) → Step 4: Score 1,000,528 customers



Results: how well it works and what drives it

Tested on a new period (Dec 2024)

0.674 AUC

WoE scorecard

0.661 AUC

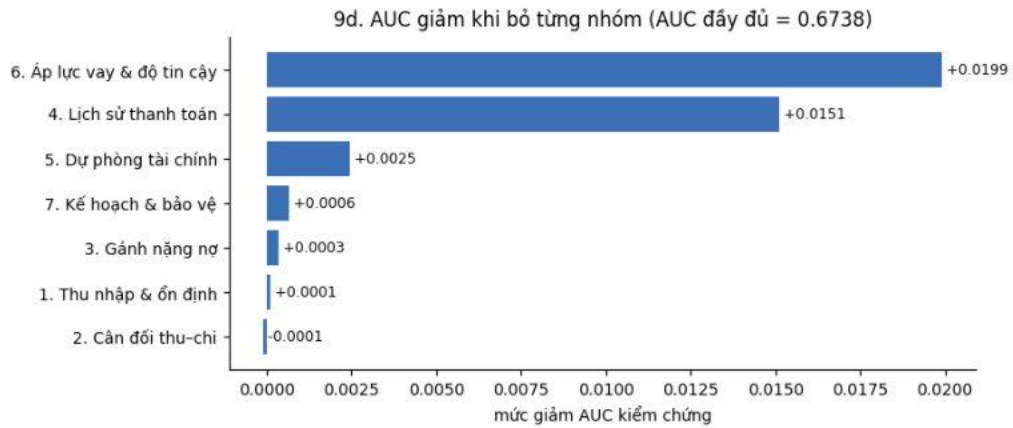
Training

0.761 AUC

Risk bands

1% → 6%

Healthiest — Highest risk

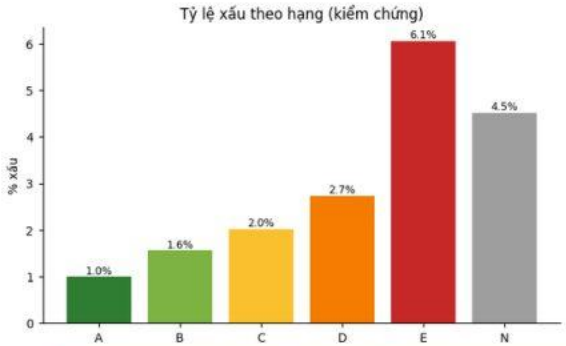


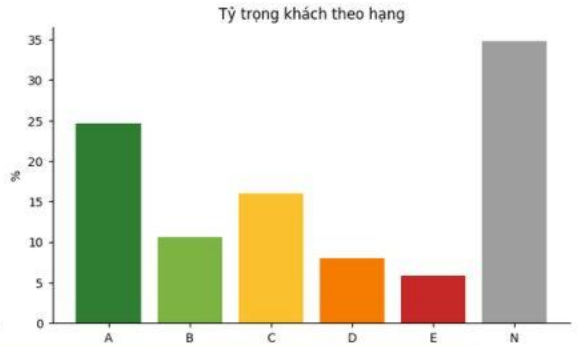
It ranks customers by repayment risk better than a traditional scorecard (AUC 0.674 vs 0.661), and the late rate rises steadily from 1% in the healthiest group to 6% in the riskiest. Accuracy is lower than in training (0.761) because the December 2024 customer pool is larger and has a higher late rate, so about 0.67 is the realistic level on new data. The score reflects credit behavior (lender trust and repayment history) rather than income, so it is best used to prioritise customers, not to predict one individual's outcome.

Customer Financial Health Score

Each customer gets a 0–100 score, a risk band from A to E, and the reasons behind it

THE BANDS RANK RISK IN THE RIGHT ORDER: late rate rises from 1% (A) to 6% (E)

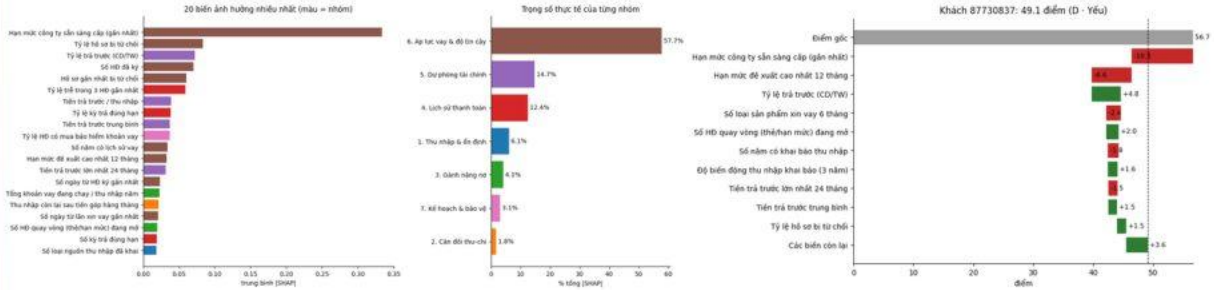




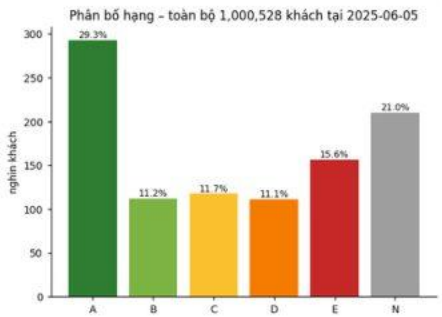
Band	Bad rate	Share of customers
A Very healthy	0,99%	2,47%
B Healthy	1,56%	1,06%
C Average	2,01%	1,60%
D Weak	2,74%	0,80%
E High risk	6,05%	0,59%
N Thin file	4,52%	3,48%

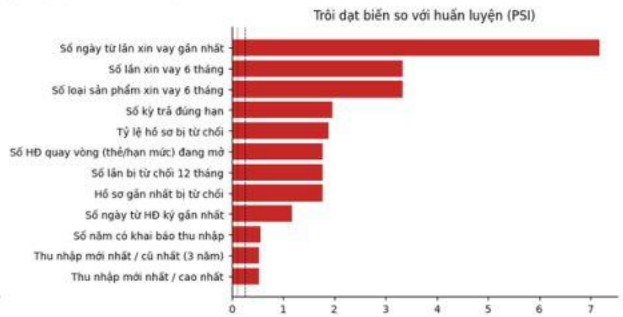
Band N is the largest group with a 4.5% bad rate, close to E: thin-file customers need their own path.

WHAT DRIVES THE SCORE: The lender's own view and repayment history drive about 70% of the score



PORTFOLIO TODAY: At 05/06/2025, 15.6% of customers are in E and 21.0% are thin file





- Top driver: the lender's offered limit, then rejection ratio, down payment ratio and number of contracts signed.
- By group: lender trust 58%, financial buffer 15%, repayment history 12%; income and debt about 10% together.
- Low offered limit removes 10.3 points, a good down payment adds 4.8, so every score comes with reasons.

The score leans on the lender's own past decisions: useful for ranking, but read it as credit behavior, not as independent proof of capacity.

Customer Engagement Score

Engagement is scored from behaviour only: 6 of 23 candidates, five aspects, one 0–100 scale

Define outcome

- Outcome: signed a new loan or became a lead deal by Jun 2025 (10.96% base rate)
- T0 = 01/02/2025, re-test T1 = 01/04/2025
- Random 50/50: calibration 500K, holdout 500K

Map factors

- 23 behavioural candidates
- Grouped into 5 groups: transactions & products, digital channels, marketing/CS feedback, application journey, churn signals

Select

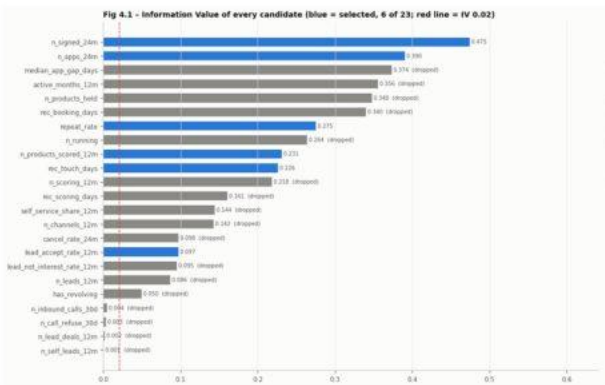
- IV ≥ 0.02, WoE correlation ≤ 0.7
- Positive weights only
- 6 of 23 kept

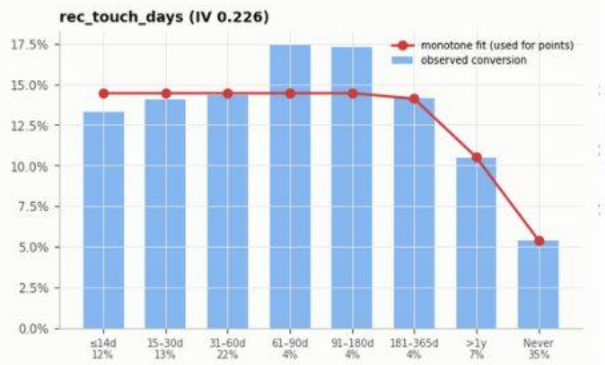
Monotone fit

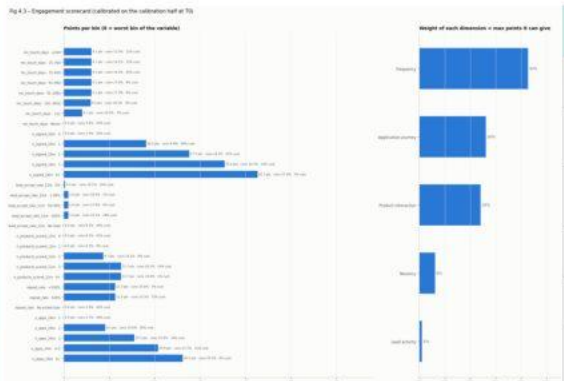
- Isotonic fit per variable
- More activity never lowers the score
- Fixes the Tết distortion in Recency

Score

- WoE points summed, rescaled to 0 –100
- Cut-offs 81.9 / 48.3 / 15.4
- 4 segments







Aspect	Variable	Conversion: low → high	Weight
Frequency	Signed loans (24m)	3.5% → 27.4%	**42,7**
Application journey	Applications (24m)	5.7% → 29.3%	**26,2**
Product	Products scored, repeat rate	8.1% → 20.2%	**24,0**
Recency	Days since last touch	5.4% → 14–17%	6,1
Lead	Lead accept rate	Weak signal	1,0
Interaction	None kept	Call answer rate is selection bias	0,0

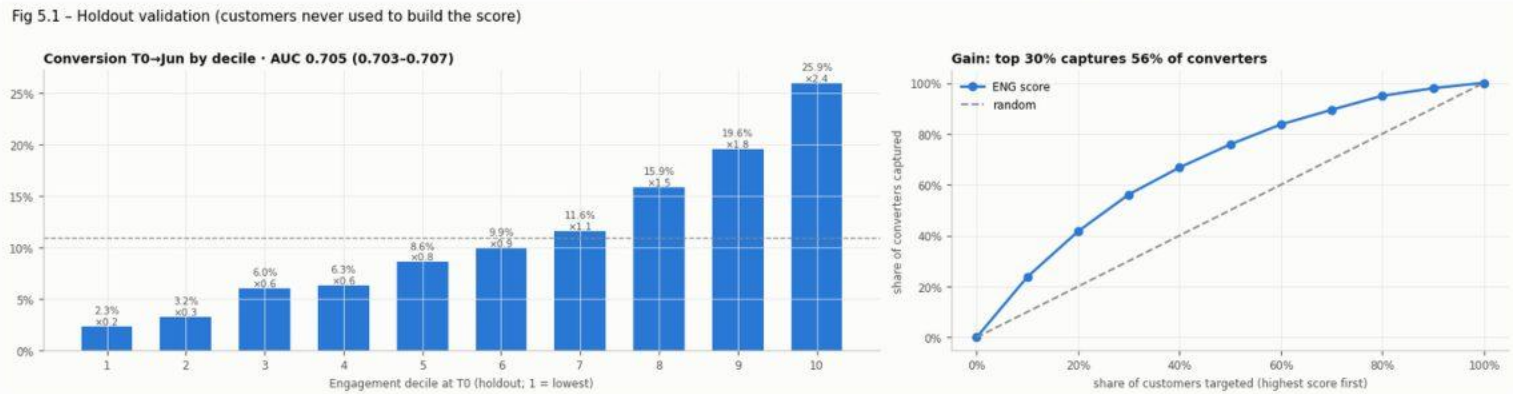
- Loans signed (IV 0.475) and applications (0.390) lead. Look-alike variables (active months, products held, booking recency) are dropped to avoid double counting
- Raw conversion peaks at 61–180 days because all customers applied around Tết. The monotone fit keeps the real pattern: never touched 5.4% vs 14–17%.

Loans signed and applications drive 69% of the score, so engagement here means how often a customer signs and applies. Interaction carries no weight because call answer rate reflects who sales chose to call, not customer intent. Customers with 4+ signed loans convert at 27.4% vs 3.5% for those with none, so loan history is the strongest signal for ranking cross-sell targets.

Customer Engagement Score

Validated on 500K unseen customers and at a later date: AUC 0.705, top decile converts 11× the bottom

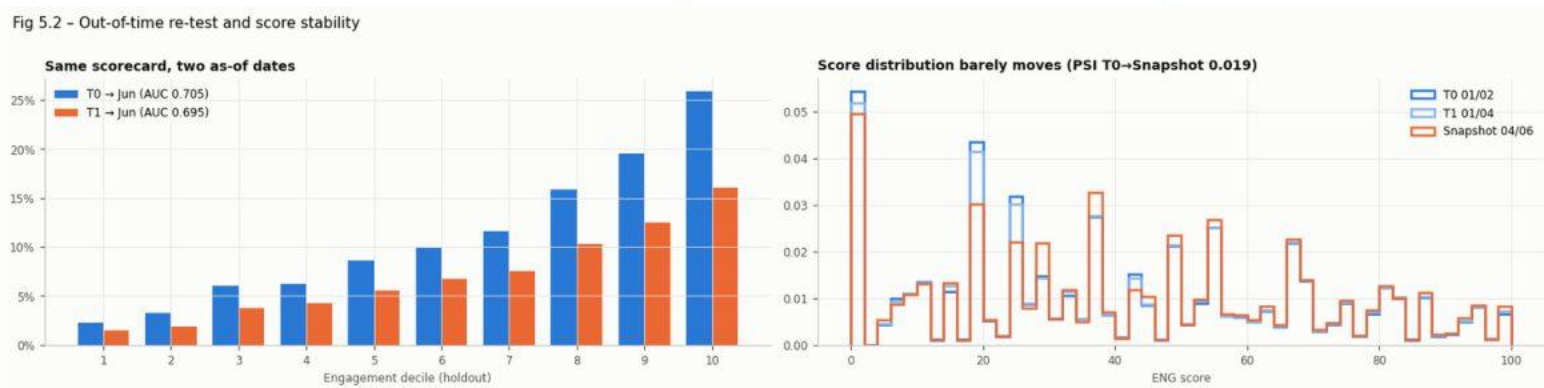
Ranks customers well on unseen data



AUC	Gini	KS
0.705	0.41	0.304

- Top decile converts at 25.9% (×2.4) vs 2.3% (×0.2) in the bottom decile, an 11× spread
- Targeting the top 30% of customers captures 56% of all converters

Holds up over time and beats simple alternatives



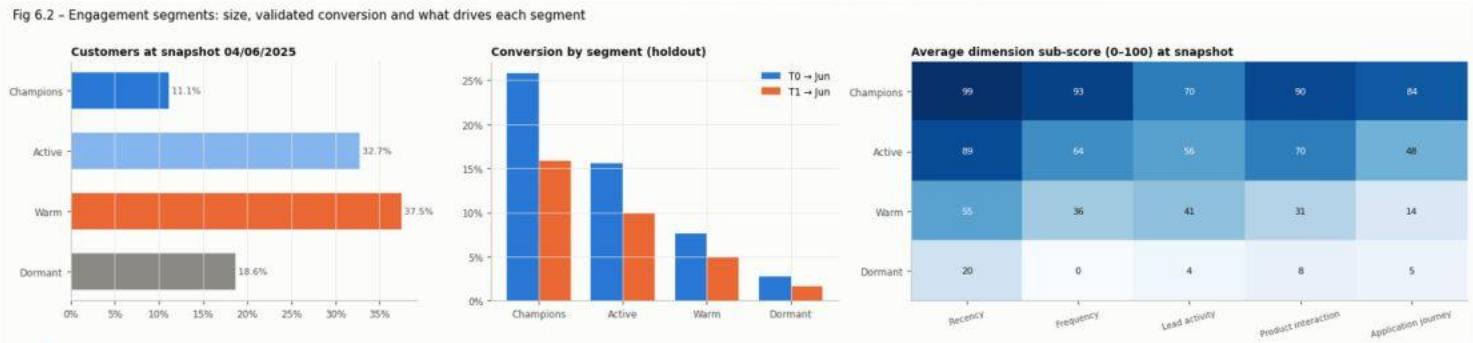
Model	AUC
RFM	0,660
Equal weights	0,688
ENG scorecard	**0,705**
Gradient boosting	0,730

The simple scorecard keeps about 97% of gradient-boosting power (0.705 vs 0.730) and beats RFM by 0.045 AUC, while every point stays explainable. Variables tied to recency drift fast (PSI 1.10 for days since last touch), so the cut-offs should be recalibrated quarterly.

Customer Engagement Score

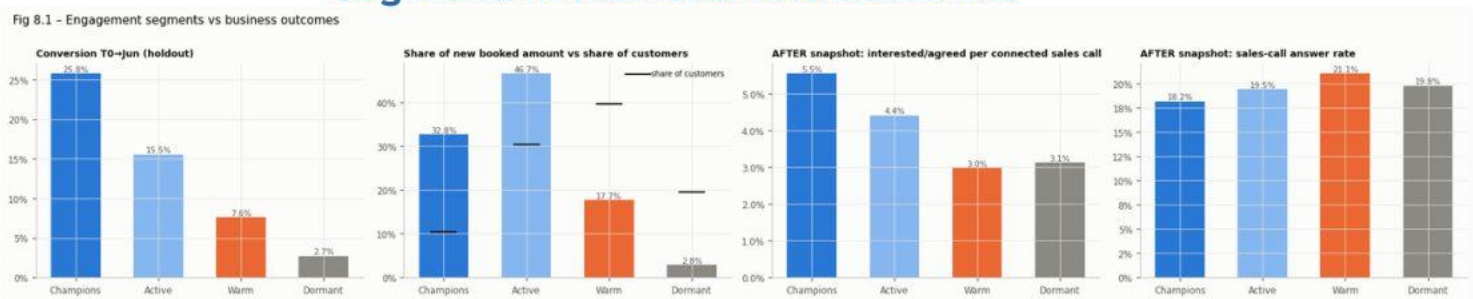
Prioritise Champions and Active: they are 44% of customers but bring 79% of new booked value

Four segments, validated on real conversion



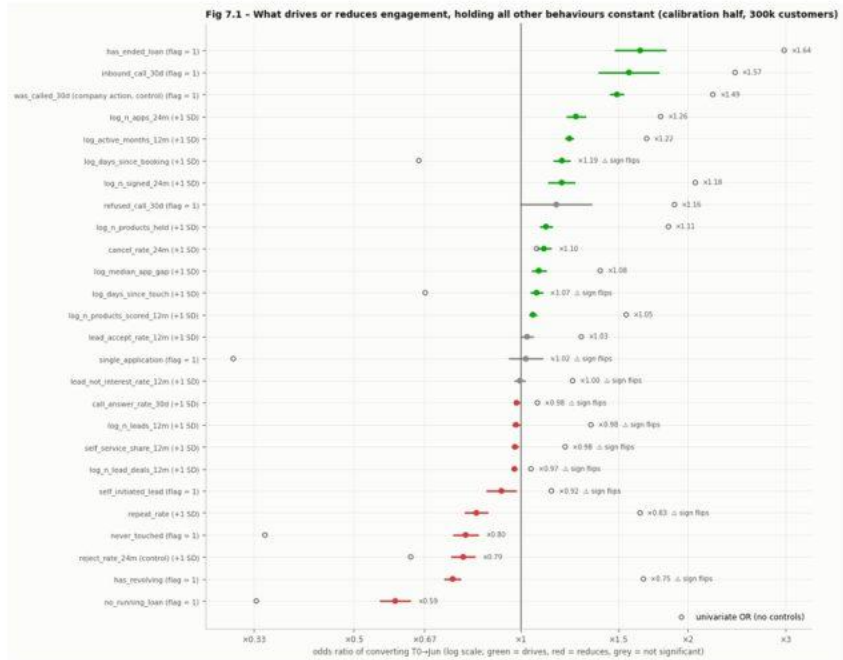
- Champions (11.1%) convert at 25.8% vs Dormant (18.6%) at 2.7%, and the order is the same at T1 (15.9% vs 1.6%)
- Warm is the largest group (37.5%): weakest on Application journey (14/100) and Product interaction (31/100)

Segments match business outcomes



- Champions + Active are 43.8% of customers but 79.5% of new booked value; Dormant is 18.6% of customers and 2.8% of value
- After the snapshot, a connected sales call ends in interest 5.5% for Champions vs 3.0–3.1% for Warm and Dormant; answer rate is flat at 18–21%

What drives or reduces engagement



The strongest lever is timing: customers whose loan has just ended are 1.64× more likely to take a new one, while customers with no running loan are 0.59× as likely. Lead volume does not help, and “not interested” is not a churn signal. Focus sales effort on Champions and Active, who generate 79% of new booked value.

Customer Segmentation

Combining Financial Health × Engagement into 7 rule-based segments (6 matrix cells + 1 insufficient-history group)

Step 1 : From scores to segments

Model Chosen

Rule-based matrix: Financial Health × Engagement

Input variables

- **Financial Health (Challenge 1): FH score 0–100 and band A–E / N**
  - High = A + B (≥ 58)
  - Mid = C + D (40–58)
  - Low = E (< 40)
  - N = no loan with ≥ 3 instalments yet
- **Engagement (Challenge 2): ENG score and Champions / Active / Warm / Dormant**
  - High = Champions + Active (≥ 56.8)
  - Low = Warm + Dormant
- **No demographics (age, gender, address) used**

THRESHOLDS

REUSED FROM CHALLENGE 1 & 2, NO NEW PARAMETERS (NO K, NO SEEDS)

Step 2. Check two scores add information?

- Spearman ρ = 0.49: related, but not redundant
- 35% of FH A customers are low-engagement (Warm + Dormant)
- 43% of FH E customers are high-engagement (Champions + Active)
- 94% of Insufficient History (N) customers are Dormant
- FH answers should we lend?; ENG answers does the customer want to borrow, when and through which channel?

Step 3. Collapse each score into tiers (thresholds reused, none new)

Axis	Tier	Rule
FH	High	A + B (≥ 58)
FH	Mid	C + D (40–58)
FH	Low	E (< 40)
FH	Insufficient History	N: no loan, incl. BNPL & card, with ≥ 3 instalments
ENG	High	Champions + Active (≥ 56.8)
ENG	Low	Warm + Dormant

Recommendation

From scoring to action: optimizing next best action via casual rationale

THE CHALLENGE

UNIFORM ACTIONS
across the entire customer portfolio leads to two major operational risks

Resource Waste: Over-allocating human operations to customers naturally maintaining healthy financial habits and pay on time, causing operational overload

Adverse Impact: Misapplying aggressive outreach or inappropriate loan offers to financially stressed / overdue customers, increasing churn and default risks.

STRATEGIC OBJECTIVE

MERE PREDICTION
"Who has a high score?"

SHIFT

CAUSAL LOGIC
"Which customer segments will actually improve their repayment behavior and generate new loan contracts when provided with the tailored Next Best Action?"

Segment	Profile & Status	Next Best Action	Strategic Business Objective
Stars (27%)	High Health (FH ≥ 58, 1.2% ever overdue) & High Engagement (actively	**Proactive upsell / cross-sell**: pre-approved top-up, CL / CC / BNPL. Renewal offer when	Maximize customer lifetime value from the highest-quality base: **452 bn VND**
Sleeping Gems (16%)	High Health (lowest risk, 0.3% ever overdue) & Low Engagement (inactive /	**Digital re-activation**: time-limited pre-approved offer (CL 51%) via app push / SMS. No	Re-activate silent, high-quality customers at minimal cost: **115 bn VND (19%)**
Growth (10%)	Moderate Health (FH 40–58, 3.4% ever overdue) & High Engagement (clear credit need)	**Right-sized offer**: small limit, down payment (BNPL / CD / CC). Telesales + app	Convert active demand into lending under controlled risk: **48 bn VND (8%)**
Cool Prospects (13%)	Moderate Health & Low Engagement (no need signal, 3.8% conversion)	**Low-cost nurture**: content and app onboarding; **no credit push** until a need signal appears	Keep warm cheaply; move to Growth when engagement rises
Stressed Seekers (7%)	Low Health (FH < 40, 26.8% ever overdue) & High Engagement (still seeking credit)	**Risk containment & financial relief**: no new credit, counselling, on-time repayment reminders	Mitigate NPL risk, protect brand reputation, support long-term recovery
At-Risk Dormant (9%)	Low Health (27% ever overdue) & Low Engagement (disengaged)	**Monitor only**: early-warning tracking, stop marketing	Protect the portfolio, avoid wasted marketing spend
Insufficient History (19%)	No repayment track record (98% never borrowed); youngest group	**Starter credit**: small BNPL / CD limit, step up after each on-time instalment. App / SMS	Build repayment history and grow the future healthy customer base

STATISTICS + A/B TESTING SETUP

For the 10% control group, the safety metric is the delinquency rate.

MODEL LAYER · SCORING & SEGMENTATION

Combines the Financial Health Score (0-100) and the Engagement Score into a transparent, explainable framework to rank and segment the portfolio into actionable business buckets.

KEY TAKEAWAY

"Not all customers should be treated equally. By clearly differentiating between on-time borrowers (High Health) and overdue borrowers (Low Health), and pairing this with causal impact validation, the business optimizes operational expenses, delivers the right action to the right person at the right time, and protects the overall health of the lending portfolio."


THANK
YOU